

Data Analytics Capstone

Multi-Modal MRI-Based Glioma Tumor Classification Using Machine
Learning and Deep Learning

Interim Report

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Introduction

Background and Context

Glioma is one of the most common and aggressive forms of brain tumors affecting the central nervous system. Early and accurate glioma detection is critically important because delayed diagnosis can significantly reduce treatment effectiveness and patient survival rates. Magnetic Resonance Imaging (MRI) is widely used in clinical practice for brain tumor diagnosis because it provides detailed visualization of brain tissues without exposing patients to ionizing radiation. Different MRI modalities such as T1, T1 Contrast-Enhanced (T1CE), T2, and FLAIR provide complementary information regarding tumor structure, edema, necrosis, and tissue abnormalities.

Traditionally, glioma diagnosis and interpretation of MRI scans are performed manually by radiologists and medical experts. However, manual interpretation is time-consuming, subjective, and prone to inter-observer variability, particularly when dealing with large-scale imaging datasets. With the increasing availability of medical imaging data, there is growing interest in applying machine learning and deep learning techniques to automate glioma tumor classification and improve diagnostic support systems. Automated classification systems can help clinicians identify tumors more efficiently and consistently while reducing workload in healthcare environments.

Recent advances in artificial intelligence (AI), machine learning (ML), and deep learning (DL) have demonstrated strong potential in medical image analysis. Machine learning models can classify tumors using engineered statistical features extracted from MRI images, while deep

learning models can automatically learn visual patterns directly from MRI slices through transfer learning architectures such as ResNet18 and EfficientNet-B0. Multi-modal MRI analysis has also become increasingly important because combining information from multiple MRI modalities can improve tumor characterization and classification performance.

The operational relevance of this study is significant for healthcare institutions, radiologists, medical imaging researchers, and AI-assisted diagnostic systems. Accurate glioma classification systems can support clinical decision-making, improve diagnostic efficiency, reduce diagnostic errors, and contribute to personalized treatment planning. The study also contributes to ongoing research in medical imaging analytics and explainable AI for healthcare applications.

Problem Statement

Despite advances in medical imaging technology, accurate glioma tumor classification remains a challenging problem due to the complexity and heterogeneity of MRI brain tumor patterns. Manual MRI interpretation by radiologists is labor-intensive and may produce inconsistent results across different experts. Although deep learning approaches have shown promise in medical imaging, their performance often depends on large labeled datasets and extensive computational resources. Furthermore, there remains limited comparative research evaluating whether engineered statistical MRI features combined with traditional machine learning models can outperform transfer learning-based deep learning approaches for glioma tumor classification. This study addresses this gap by comparing machine learning and deep

learning models using multi-modal MRI data to determine the most effective approach for accurate glioma tumor classification.

Purpose of the Study

The purpose of this study is to develop and evaluate machine learning and deep learning models for glioma tumor classification using multi-modal MRI data. The study aims to:

- classify tumor and non-tumor MRI slices,
- compare the performance of machine learning and deep learning approaches,
- evaluate the effectiveness of multi-modal MRI integration,
- analyze the contribution of engineered MRI statistical features,
- and identify the best-performing classification model for glioma tumor detection.

The study specifically focuses on:

- feature extraction,
- classification,
- transfer learning,
- comparative model evaluation,
- and performance analysis using metrics such as accuracy, precision, recall, F1-

score, and ROC-AUC.

Interim Project Status (Progress Snapshot)

Completed

- Collected and extracted the UTSW-Glioma MRI dataset from TCIA
- Performed MRI preprocessing and dataset organization
- Converted MRI images from NIfTI format
- Applied tumor segmentation and MRI slice extraction
- Extracted statistical intensity-based MRI features
- Completed data cleaning and preprocessing
- Performed exploratory data analysis (EDA)
- Conducted feature scaling and normalization
- Trained multiple machine learning models:
 - Logistic Regression
 - SVM
 - KNN
 - Random Forest
 - Gradient Boosting
 - XGBoost
- Implemented deep learning transfer learning models:
 - ResNet18
 - EfficientNet-B0
- Generated evaluation metrics and visualizations
- Performed machine learning vs deep learning comparison
- Conducted hypothesis validation analysis

- Generated preliminary performance evaluation results

In Progress

- Additional feature engineering optimization
- Hyperparameter tuning for selected models
- Comparative validation analysis
- Final report writing and interpretation refinement
- Preparation of GitHub repository documentation
- Figure and table formatting for final dissertation submission

Pending

- Final model optimization and tuning
- Extended validation experiments
- Additional literature integration
- Final conclusion and recommendations
- Appendix preparation and code documentation
- Final dissertation submission preparation

Scope and Objectives

The scope of this study focuses on glioma tumor classification using multi-modal MRI data and artificial intelligence-based classification approaches. The research evaluates both machine learning models using engineered MRI statistical features and deep learning transfer learning models trained directly on MRI image slices.

The primary objectives of the study are:

1. To evaluate the effectiveness of machine learning models for glioma tumor classification using engineered MRI statistical features.
2. To evaluate the performance of transfer learning-based deep learning models for MRI image classification.
3. To analyze the impact of multi-modal MRI integration on classification performance.
4. To compare machine learning and deep learning approaches for glioma tumor detection and classification.

Research Questions

Research Question 1 (RQ1)

How effectively can machine learning models classify glioma tumors using statistical intensity-based features extracted from multi-modal MRI scans?

Research Question 2 (RQ2)

How accurately can deep learning transfer learning models classify glioma tumors directly from MRI image slices without manual feature extraction?

Research Question 3 (RQ3)

Does integrating multiple MRI modalities improve glioma tumor classification performance compared to using a single MRI modality?

Research Question 4 (RQ4)

Which approach provides better glioma tumor classification performance: traditional machine learning models using engineered MRI features or deep learning models trained directly on MRI images?

Literature survey

Article Summary 1

Edoardo Agosti et al. (2024) conducted a systematic review to investigate the role of glioma stem cells (GSCs) in glioma progression, therapy resistance, and tumor recurrence. Using PRISMA guidelines, the authors analyzed 65 studies from PubMed, MEDLINE, and EMBASE focusing on molecular signaling pathways such as Notch, STAT3, Wnt/ β -catenin, HIF, and PI3K/AKT.

The study found that glioma stem cells significantly contribute to tumor growth, invasion, and treatment resistance, while targeted therapies against these pathways may improve glioblastoma outcomes. This article supports the current research by emphasizing tumor heterogeneity, the importance of multi-modal analysis, and the need for advanced machine learning and deep learning models for accurate glioma classification using MRI data.

Article Summary 2

Isabel Arrillaga-Romany et al. (2024) evaluated the effectiveness of ONC201 (dordaviprone) as a targeted therapy for recurrent H3 K27M–mutant diffuse midline glioma (DMG), an aggressive brain tumor with poor prognosis. The study analyzed 50 patients across multiple clinical trials using MRI-based tumor assessment and Response Assessment in Neuro-Oncology (RANO) criteria.

The findings showed that ONC201 demonstrated clinically meaningful tumor response, prolonged survival outcomes, and acceptable safety with limited severe adverse effects. This study supports the current research by emphasizing the clinical importance of accurate MRI-based glioma evaluation, tumor classification, and advanced predictive analysis for improving glioma diagnosis, treatment monitoring, and therapeutic decision-making.

Article Summary 3

Tianzhen Hua et al. (2022) reviewed the role of glioma-neuronal interactions in glioma progression, focusing on how neuronal signaling, neurotransmitters, ion channels, and tumor microenvironment communication influence tumor growth and invasion. The review analyzed molecular mechanisms, therapeutic targets, and emerging technologies related to glioma progression and treatment.

The study found that neuron-glioma communication significantly promotes glioma proliferation, invasion, therapy resistance, and tumor recurrence through pathways involving glutamate, GABA, NLGN3, PI3K/mTOR, and ion-channel signaling. This article supports the current research by emphasizing glioma heterogeneity, MRI-relevant tumor complexity, and the importance of advanced computational and multi-modal approaches for accurate glioma classification and predictive modeling.

Article Summary 4

Leon Jekel et al. (2022) conducted a systematic review of machine learning applications for differentiating glioma from brain metastasis using MRI imaging data. The study reviewed 29

research articles and evaluated machine learning models, imaging modalities, validation methods, and reporting quality for glioma classification tasks.

The review found that machine learning models, particularly Support Vector Machine (SVM) and deep learning approaches, achieved promising classification performance with pooled AUC values above 0.90. However, the study highlighted major limitations including small datasets, lack of external validation, class imbalance, and insufficient reporting quality, emphasizing the need for robust multi-center validation and standardized AI methodologies in glioma classification research.

Article Summary 5

Anuj Kumar et al. (2023) investigated the use of machine-learning-based radiomics features extracted from MRI images for classifying low-grade and high-grade gliomas. The study used T2-weighted MRI scans from 83 histopathologically confirmed glioma patients and applied multiple machine learning classifiers including Random Forest, SVM, Gradient Boost, Naive Bayes, and AdaBoost.

The findings showed that the Random Forest classifier achieved the best performance with an accuracy of 83% and AUC of 0.81, while SVM also demonstrated strong classification capability. The study supports the current research by validating the effectiveness of radiomics-based machine learning models, MRI feature extraction, and comparative classifier evaluation for accurate glioma grading and non-invasive tumor classification.

Article Summary 6

Sara Merkaj et al. (2022) reviewed machine learning tools used for image-based glioma grading and examined the reporting quality, reproducibility, and clinical applicability of these AI models. The study discussed commonly used machine learning and deep learning algorithms, including SVM and CNN, while analyzing challenges related to datasets, external validation, reporting standards, and clinical implementation.

The review found that most glioma grading studies showed poor reporting quality with low TRIPOD adherence and high risk of bias, despite achieving high classification accuracies using MRI-based ML and DL approaches. The study supports the current research by emphasizing the importance of standardized validation, multi-modal MRI analysis, reproducibility, and comparative evaluation of machine learning and deep learning models for reliable glioma classification.

Article Summary 7

Takanori Ohnishi (2024) reviewed the current status and future directions of glioma invasion research, focusing on the molecular mechanisms, tumor microenvironment (TME), imaging techniques, and therapeutic strategies associated with glioblastoma invasion. The study discussed how hypoxia, extracellular matrix interactions, glioma stem-like cells, and MRI/PET imaging contribute to tumor progression and invasive behavior.

The review found that glioma invasion is strongly influenced by tumor microenvironment interactions, hypoxia-related signaling pathways, and glioma stem-like cells, while advanced imaging methods such as MRI, MR spectroscopy, and PET imaging can improve detection of infiltrative tumor regions. This article supports the current research by reinforcing the

importance of multi-modal MRI analysis, tumor heterogeneity assessment, and advanced AI-based classification methods for accurate glioma diagnosis and prediction.

Article Summary 8

Sho Tamai et al. (2022) reviewed the role of the tumor microenvironment (TME) in glioma invasion, focusing on hypoxia, extracellular matrix remodeling, astrocytes, microglia, glioma stem cells, and therapy-induced microenvironmental changes. The study analyzed molecular signaling pathways such as HIF-1 α , JAK/STAT, NF- κ B, PI3K/Akt, and Notch signaling associated with glioma progression and invasion.

The review found that hypoxia, extracellular matrix interactions, tumor-associated astrocytes, and glioma-associated macrophages significantly promote glioma invasion, treatment resistance, and tumor heterogeneity. This article supports the current research by emphasizing the complexity of glioma behavior, the importance of multi-modal MRI analysis, and the need for advanced machine learning and deep learning techniques for accurate glioma classification and prediction.

Article Summary 9

Natalia Vasileva et al. (2024) investigated the effectiveness of the recombinant oncolytic vaccinia virus VV-GMCSF-Lact combined with temozolomide (TMZ) for glioma treatment using immortalized and patient-derived glioma cell cultures and U87 MG xenograft models. The study evaluated cytotoxicity, tumor growth inhibition, histological changes, and treatment sequencing strategies for improving glioblastoma therapy.

The findings showed that VV-GMCSF-Lact demonstrated stronger antitumor activity than TMZ alone, while combination therapy significantly improved tumor destruction and growth inhibition in glioblastoma models. This article supports the current research by emphasizing glioma heterogeneity, MRI-relevant tumor complexity, and the importance of advanced predictive and AI-based approaches for improving glioma diagnosis, treatment evaluation, and personalized therapeutic strategies.

Article Summary 10

Rachel Zhao et al. (2022) investigated machine learning–based survival prediction in glioma patients using large-scale registry data from 3,462 patients diagnosed between 2000 and 2018. The study compared Cox Proportional Hazards (CPH), Support Vector Machine (SVM), and Random Forest (RF) models using clinical and treatment-related variables such as age, tumor histology, surgery, chemotherapy, and radiation therapy.

The findings showed that SVM achieved the highest survival prediction performance with a concordance index of 0.771, while chemotherapy and radiation therapy emerged as the most important predictive features. This article supports the current research by validating the effectiveness of machine learning models for glioma prediction, comparative model evaluation, and the importance of integrating clinical and MRI-related features for improved glioma prognosis and personalized treatment planning.

Table A

Literature relevance matrix

Author (Year)	Domain/Context	Dataset/Setting	Method(s)	Key Findings	How it supports RQ(s) / project decisions
Agosti et al. (2024)	Glioma stem cell biology and glioblastoma progression	Systematic review of 65 glioma studies from PubMed, MEDLINE, and EMBASE	PRISMA-based systematic literature review and molecular pathway analysis	Identified critical glioma stem cell pathways including Notch, STAT3, Wnt/ β -catenin, HIF, and PI3K/AKT involved in tumor progression, therapy resistance, and recurrence	Supports RQ1 and RQ3 by emphasizing glioma heterogeneity and the importance of multi-factor tumor analysis. Supports RQ4 by highlighting

					g the need for advanced computatio nal approaches for accurate glioma diagnosis and classificatio n.
Arrilla ga- Roman y et al. (2024)	Recurrent H3 K27M–mutant diffuse midline glioma treatment and MRI-based tumor	Clinical trials involving 50 recurrent diffuse midline glioma patients	Clinical trial analysis, MRI evaluation, RANO- HGG/LGG assessment, survival and	ONC201 showed durable tumor response, improved survival outcomes, and	Supports RQ1 and RQ4 by emphasizin g the importance of accurate

	evaluation		response analysis	acceptable safety in recurrent glioma patients	glioma classification and MRI- based tumor assessment for treatment evaluation and clinical decision- making.
Hua et al. (2022)	Glioma- neuronal interactions and tumor progression mechanisms	Review of experimental glioma studies, molecular signaling pathways, and therapeutic investigations	Literature review of neuron-glioma signaling, molecular pathways, and therapeutic strategies	Identified that neuronal signaling, neurotransmitt ers, ion channels, and tumor microenvironm	Supports RQ1, RQ3, and RQ4 by highlightin g glioma complexity, tumor

				ent interactions significantly contribute to glioma progression, invasion, and therapy resistance	heterogenei ty, and the need for advanced multi- modal MRI analysis and AI- based classificatio n approaches for improved glioma detection and prediction.
Jekel et al.	Machine learning–	Systematic review of 29	PRISMA systematic	Machine learning and	Supports RQ1, RQ3,

(2022)	based differentiation of glioma and brain metastasis using MRI	MRI-based machine learning studies	review, meta-analysis, TRIPOD reporting evaluation, ML/DL performance comparison	deep learning models achieved high classification performance, but limitations included lack of external validation, small datasets, and poor reporting standards	and RQ4 by validating the effectiveness of AI-based MRI classification approaches and emphasizing the importance of multi-modal MRI integration, robust validation, and
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					comparative ML vs DL analysis in glioma research.
Kumar et al. (2023)	MRI radiomics and machine learning for glioma grading	83 histopathologically confirmed glioma patients using T2-weighted MRI	Radiomics feature extraction, Recursive Feature Elimination (RFE), Random Forest, SVM, Gradient Boost, Naive Bayes, and AdaBoost classifiers	Random Forest achieved the highest performance with 83% accuracy and AUC of 0.81, demonstrating the effectiveness of radiomics-based MRI classification	Supports RQ1, RQ2, and RQ3 by validating MRI-based radiomics, feature engineering, and machine learning approaches for accurate

					glioma grading and comparativ e model evaluation.
Merkaj et al. (2022)	Machine learning and deep learning for image- based glioma grading	Review of over 80 MRI-based glioma grading studies using public and institutional datasets	Literature review, TRIPOD reporting analysis, ML/DL workflow evaluation, comparative algorithm analysis	SVM and CNN achieved high classification accuracy, but most studies lacked external validation and demonstrated poor reporting quality and reproducibility	Supports RQ1, RQ2, RQ3, and RQ4 by validating MRI-based ML/DL approaches , emphasizin g model comparison , external validation,

					standardize d reporting, and clinical applicabilit y for glioma classificatio n research.
Ohnishi (2024)	Glioblastoma invasion mechanisms, tumor microenviron ment, and imaging-based detection	Review of glioma invasion studies, MRI/PET imaging studies, and tumor microenviron ment research	Literature review of molecular pathways, imaging modalities, and invasion- related therapeutic approaches	Glioma invasion is regulated by hypoxia, extracellular matrix interactions, glioma stem- like cells, and tumor microenvironm	Supports RQ1, RQ3, and RQ4 by emphasizin g glioma heterogenei ty, multi- modal imaging analysis,

				ent signaling, while advanced imaging improves detection of infiltrative tumor regions	and the need for advanced predictive and classification n approaches for accurate glioma diagnosis and invasion assessment.
Tamai et al. (2022)	Tumor microenviron ment and glioma invasion	Review of glioma invasion studies, molecular	Literature review of hypoxia signaling, ECM	Hypoxia, HIF signaling, extracellular matrix remodeling,	Supports RQ1, RQ3, and RQ4 by highlightin

	mechanisms	signaling pathways, and tumor microenvironment research	remodeling, astrocyte/microglia interactions, and invasion-related pathways	and tumor-associated astrocytes/microglia strongly contribute to glioma invasion, progression, and therapy resistance	g glioma heterogeneity, complex MRI-visible tumor characteristics, and the importance of advanced AI-based multi-modal MRI classification approaches for glioma prediction
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					and diagnosis.
Vasilev a et al. (2024)	Oncolytic virotherapy and chemotherapy for glioblastoma treatment	Immortalized glioma cell lines, patient- derived glioma cultures, and U87 MG xenograft glioblastoma mouse models	Cytotoxicity assays, combination therapy analysis, histological evaluation, tumor growth inhibition analysis, in vitro and in vivo experiments	VV-GMCSF- Lact demonstrated stronger antitumor efficacy than TMZ alone, while combination therapy significantly improved tumor destruction and treatment effectiveness	Supports RQ1, RQ3, and RQ4 by emphasizin g glioma heterogenei ty, personalize d treatment strategies, and the importance of advanced computatio nal and MRI-based

					approaches for glioma classification, prediction, and treatment evaluation.
Zhao et al. (2022)	Machine learning–based glioma survival prediction and prognosis analysis	BC Cancer Registry dataset containing 3,462 glioma patients diagnosed between 2000–2018	Cox Proportional Hazards (CPH), Support Vector Machine (SVM), Random Forest (RF), survival prediction	SVM achieved the best survival prediction performance (c-index 0.771), while chemotherapy and radiation therapy were identified as the most	Supports RQ1, RQ2, and RQ4 by validating machine learning approaches for glioma prediction, comparative model

			modeling, ROC and Kaplan–Meier analysis	important predictive features	evaluation, and personalize d clinical decision- making using large- scale glioma datasets and predictive modeling.
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Data Description

This study uses a publicly available multi-modal brain MRI dataset derived from the UTSW Glioma Dataset hosted by The Cancer Imaging Archive (TCIA). The dataset contains MRI scans collected from glioma patients and was developed to support brain tumor analysis, segmentation, and classification research.

The dataset includes four MRI modalities:

- T1-weighted MRI (T1),
- T1 Contrast-Enhanced MRI (T1CE),
- T2-weighted MRI (T2),
- and FLAIR MRI.

The MRI images were originally stored in **NIfTI (.nii/.nii.gz)** format and later converted into slice-based image representations for preprocessing, feature extraction, and model training.

Tumor segmentation was applied before feature extraction to isolate clinically relevant tumor regions. Statistical intensity-based features were then extracted from each MRI modality.

The dataset was used for:

- machine learning classification,
- deep learning transfer learning,
- feature engineering,
- and comparative analysis between single-modality and multi-modal MRI

classification.

Data Source(s) and Access

The dataset used in this study was obtained from:

UTSW Glioma Dataset (TCIA)

<https://www.cancerimagingarchive.net/collection/utsw-glioma/>

The dataset is publicly accessible for academic and research purposes.

Dataset Overview

The dataset contains MRI scans collected from approximately **625 glioma patients**. Multi-modal MRI imaging was used to capture complementary tumor-related information across different MRI sequences.

The following MRI modalities were included:

- T1,
- T1CE,
- T2,
- and FLAIR.

After preprocessing, cleaning, and zero-value filtering, the final structured dataset contained:

- 2,026 MRI image slices,
- 24 engineered MRI statistical features,
- 1 target label variable,
- and 1 patient identifier variable.

Table B

Dataset Summary

Attribute	Value
Number of records (rows)	2026
Number of variables (columns)	26
Time period	Public benchmark MRI dataset
Unit of analysis	MRI image slice
MRI modalities	T1, T1CE, T2, FLAIR
Target variable	label
Target definition	0 = Non-Tumor, 1 = Tumor

Table C**Data Dictionary**

Variable Name	Definition	Datatype	Allowed Values/Range	Missing Values Handling	Notes
t1_mean	Mean intensity of T1 MRI	Float	Continuous positive values	No missing values	Statistical MRI feature
t1_std	Standard deviation of T1 MRI intensities	Float	Continuous positive values	No missing values	Measures intensity variability
t1_max	Maximum intensity in T1 MRI	Float	Continuous positive values	No missing values	Bright tumor regions
t1_min	Minimum intensity in T1 MRI	Float	Continuous positive values	No missing values	Low-intensity tissue
t1_median	Median intensity in T1 MRI	Float	Continuous positive values	No missing values	Robust central tendency
t1_range	Intensity range in T1	Float	Continuous positive	No missing	Feature

	MRI		values	values	spread
t1ce_mean	Mean intensity in T1CE MRI	Float	Continuous positive values	No missing values	Contrast-enhanced tumor feature
t1ce_std	Standard deviation in T1CE MRI	Float	Continuous positive values	No missing values	Enhancement variability
t1ce_max	Maximum intensity in T1CE MRI	Float	Continuous positive values	No missing values	Bright enhancement areas
t1ce_min	Minimum intensity in T1CE MRI	Float	Continuous positive values	No missing values	Background intensity
t1ce_median	Median intensity in T1CE MRI	Float	Continuous positive values	No missing values	Stable enhancement measure
t1ce_range	Intensity range in T1CE MRI	Float	Continuous positive values	No missing values	Enhancement spread
t2_mean	Mean intensity in T2 MRI	Float	Continuous positive values	No missing values	Fluid-sensitive feature

t2_std	Standard deviation in T2 MRI	Float	Continuous positive values	No missing values	Variability feature
t2_max	Maximum intensity in T2 MRI	Float	Continuous positive values	No missing values	Bright abnormal regions
t2_min	Minimum intensity in T2 MRI	Float	Continuous positive values	No missing values	Dark tissue regions
t2_median	Median intensity in T2 MRI	Float	Continuous positive values	No missing values	Central tendency
t2_range	Intensity range in T2 MRI	Float	Continuous positive values	No missing values	MRI spread
flair_mean	Mean intensity in FLAIR MRI	Float	Continuous positive values	No missing values	Edema-sensitive feature
flair_std	Standard deviation in FLAIR MRI	Float	Continuous positive values	No missing values	Intensity variability
flair_max	Maximum intensity in	Float	Continuous positive	No missing	Bright edema

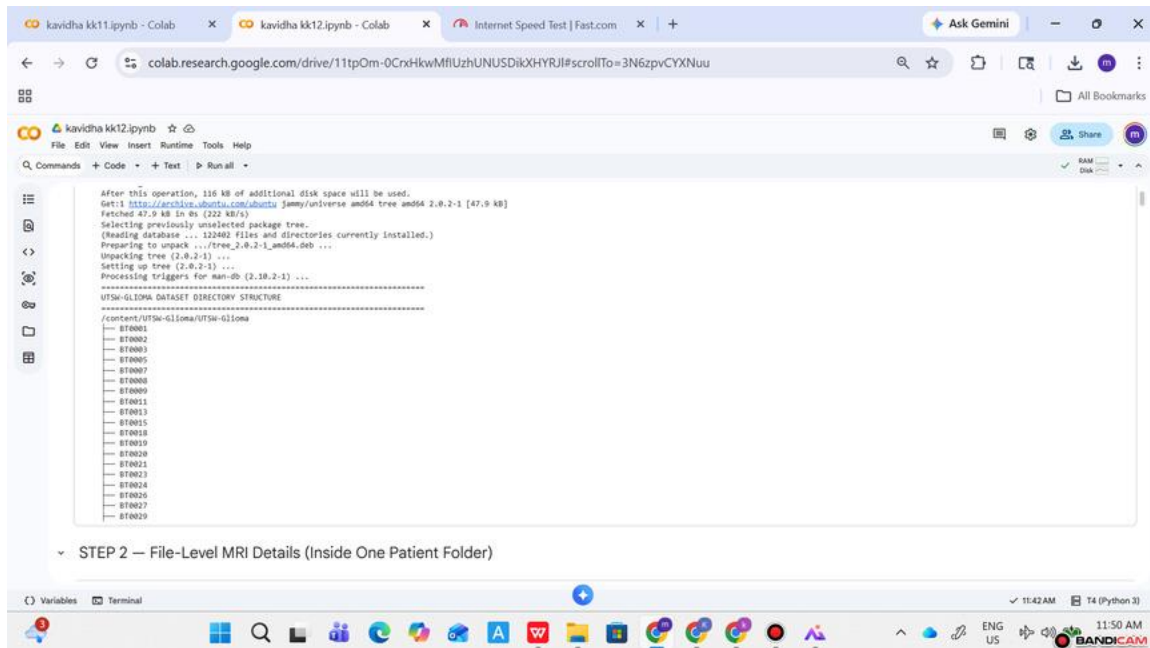
	FLAIR MRI		values	values	regions
flair_min	Minimum intensity in FLAIR MRI	Float	Continuous positive values	No missing values	Background signal
flair_median	Median intensity in FLAIR MRI	Float	Continuous positive values	No missing values	Stable edema representation
flair_range	Intensity range in FLAIR MRI	Float	Continuous positive values	No missing values	MRI heterogeneity
label	Tumor classification target	Integer	0 = Non-Tumor, 1 = Tumor	No missing values	Binary target variable
patient	Unique patient identifier	String	Alphanumeric	No missing values	Used for GroupKFold validation

GitHub Data Availability Statement

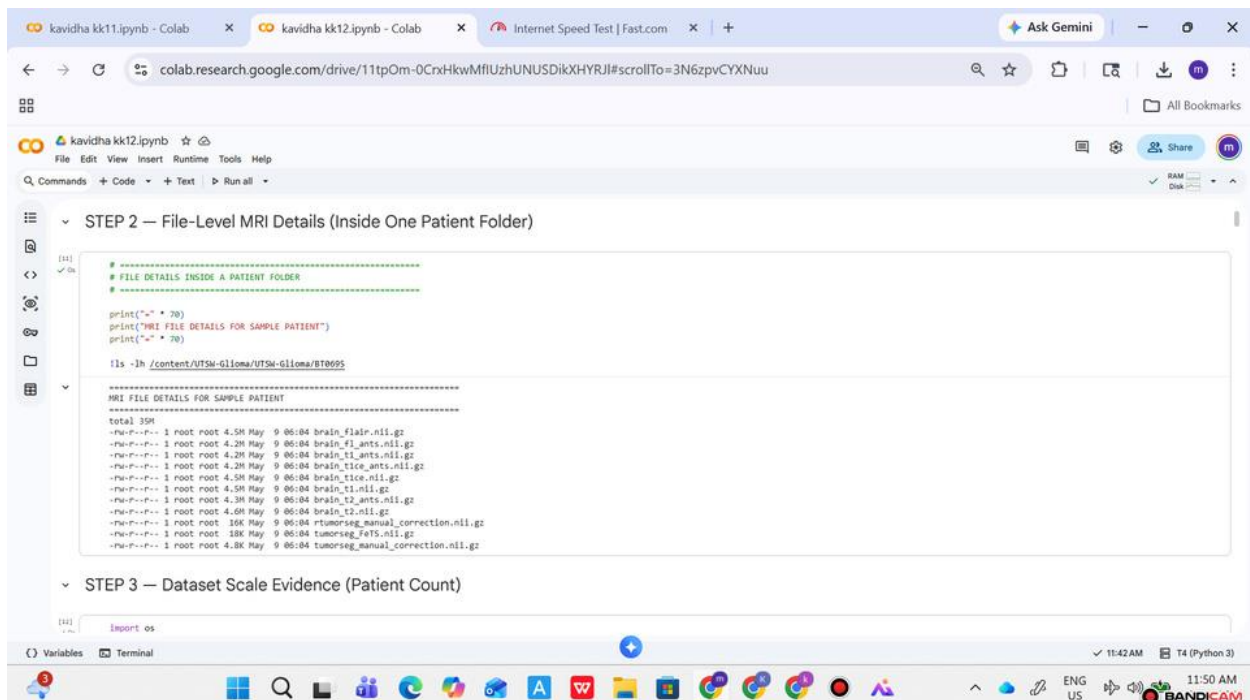
<https://github.com/kavidha123/great-learning>

Screenshots / Evidence

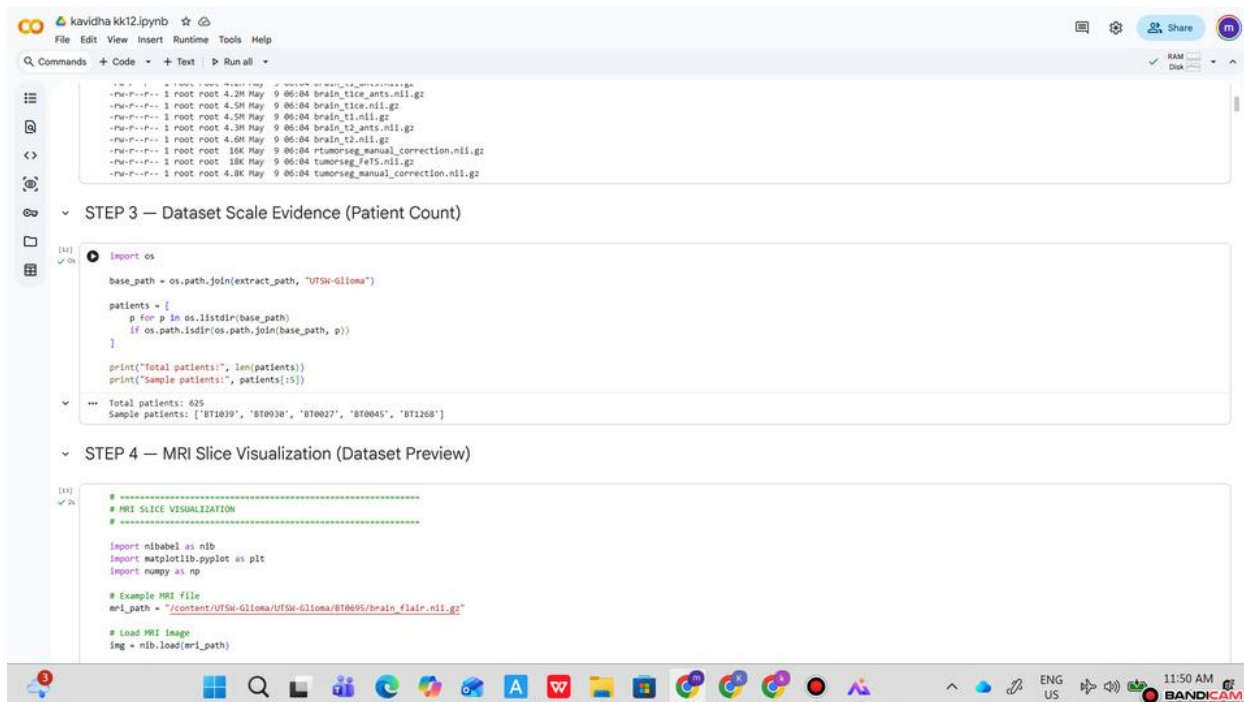
STEP 1 — Dataset Directory Tree (Hierarchy Screenshot)



STEP 2 — File-Level MRI Details (Inside One Patient Folder)



STEP 3 — Dataset Scale Evidence (Patient Count)



```
File Edit View Insert Runtime Tools Help
Commands + Code + Text + Run all +

[10/11] 1 root root 4.2M May 9 06:04 brain_t1ce_ants.nii.gz
[10/11] 1 root root 4.5M May 9 06:04 brain_t1ce.nii.gz
[10/11] 1 root root 4.5M May 9 06:04 brain_t1.nii.gz
[10/11] 1 root root 4.3M May 9 06:04 brain_t2_ants.nii.gz
[10/11] 1 root root 4.6M May 9 06:04 brain_t2.nii.gz
[10/11] 1 root root 16K May 9 06:04 rtumorseg_manual_correction.nii.gz
[10/11] 1 root root 18K May 9 06:04 tumorseg_FeTS.nii.gz
[10/11] 1 root root 4.8K May 9 06:04 tumorseg_manual_correction.nii.gz

STEP 3 — Dataset Scale Evidence (Patient Count)

[10/11] 1 Import os
base_path = os.path.join(extract_path, "UTSW-Glioma")
patients = [
    p for p in os.listdir(base_path)
    if os.path.isdir(os.path.join(base_path, p))
]
print("Total patients:", len(patients))
print("Sample patients:", patients[:5])

Total patients: 625
Sample patients: ['BT1030', 'BT0030', 'BT0027', 'BT0045', 'BT1268']

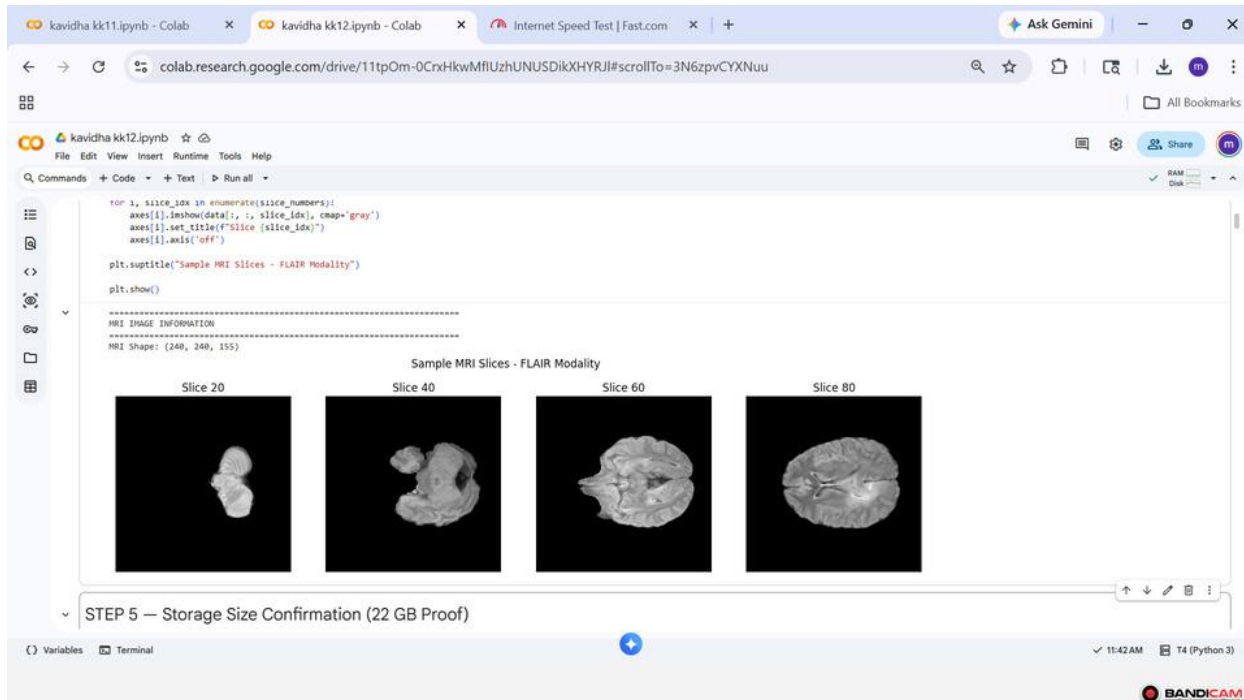
STEP 4 — MRI Slice Visualization (Dataset Preview)

[10/11] 1 # =====
# MRI SLICE VISUALIZATION
# =====
import nibabel as nib
import matplotlib.pyplot as plt
import numpy as np

# Example MRI file
mri_path = "/content/UTSW-Glioma/UTSW-Glioma/BT0045/brain_flair.nii.gz"

# Load MRI image
img = nib.load(mri_path)
```

STEP 4 — MRI Slice Visualization (Dataset Preview)



```
File Edit View Insert Runtime Tools Help
Commands + Code + Text + Run all +

[10/11] 1 for i, slice_idx in enumerate(slice_numbers):
axes[i].imshow(data[i, :, slice_idx], cmap="gray")
axes[i].set_title(f"Slice {slice_idx}")
axes[i].axis('off')

plt.suptitle("Sample MRI Slices - FLAIR Modality")
plt.show()

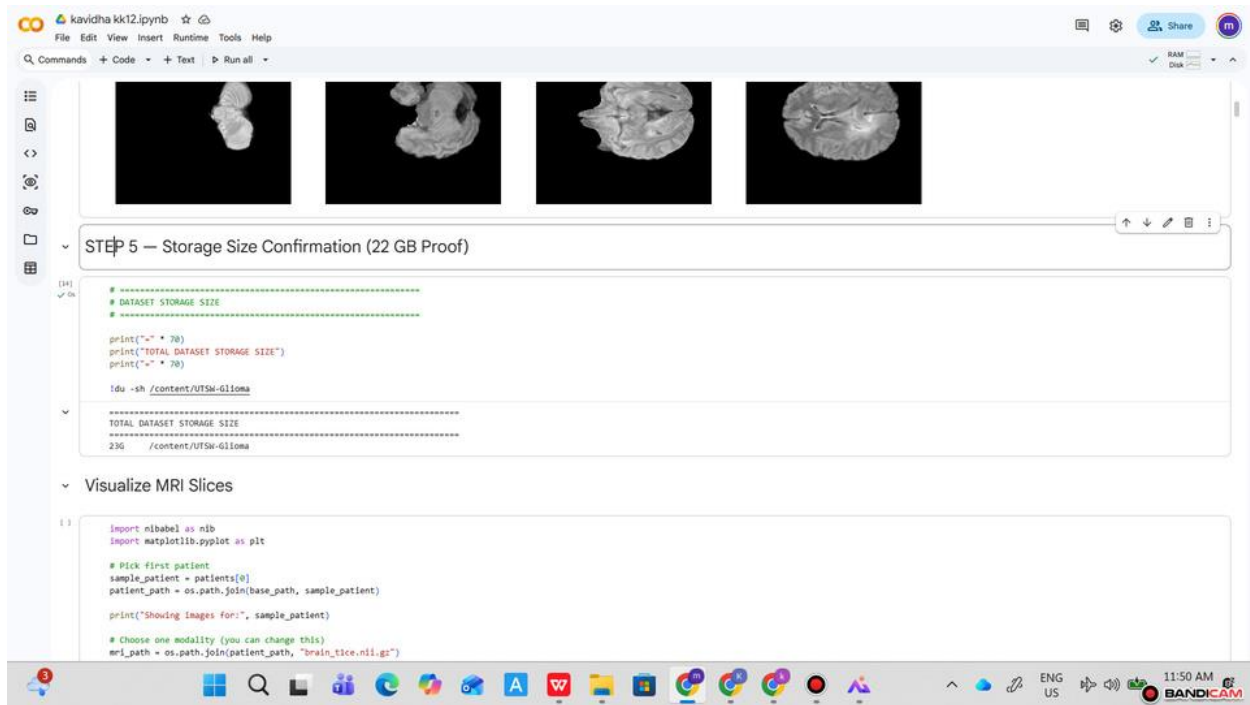
MRI IMAGE INFORMATION
MRI Shape: (240, 240, 155)

Sample MRI Slices - FLAIR Modality

Slice 20 Slice 40 Slice 60 Slice 80

STEP 5 — Storage Size Confirmation (22 GB Proof)
```

STEP 5 — Storage Size Confirmation (22 GB Proof)



Analysis

The analysis phase included:

- data cleaning,
- exploratory data analysis (EDA),
- feature engineering,
- feature scaling,
- machine learning classification,
- deep learning transfer learning,
- and comparative performance evaluation.

The primary objective of the analysis was to identify meaningful MRI intensity-based patterns associated with glioma tumors and evaluate the classification performance of different machine learning and deep learning approaches.

All tables, graphs, figures, and visualizations were interpreted to identify insights relevant to the research questions and study objectives.

Data Cleaning

Several preprocessing and cleaning operations were applied before model training to improve dataset quality and classification reliability.

The original dataset contained:

- 2,136 MRI image slice records.

After preprocessing and cleaning:

- the dataset size reduced to 2,048 records.

After zero-value filtering:

- the final cleaned dataset contained 2,026 MRI image slices.

No missing values or duplicate rows were detected after preprocessing.

Feature scaling using StandardScaler normalization transformed all MRI statistical features to:

- approximately zero mean,
- and unit variance,

which improved model stability and optimization performance.

Outlier analysis identified several extreme MRI intensity values across statistical features such as:

- t1_max,
- t1ce_max,
- t2_range,
- and flair_max.

These outliers were retained because they represented clinically meaningful tumor heterogeneity rather than data errors.

Table D

Data Cleaning Log

Issue	Variables Affected	Detection Method	Treatment Applied	Rationale
Missing values	All variables	isnull().sum() analysis	No treatment required	No missing values detected
Duplicate rows	Entire dataset	duplicated().sum()	No duplicates found	Dataset integrity maintained
Zero-value MRI slices	MRI intensity features	Zero-value filtering	Invalid MRI slices removed	Remove non-informative MRI records
Outliers	MRI intensity features	Statistical summary, IQR analysis, boxplots	Retained after verification	Represent clinically meaningful tumor variability
Feature scale imbalance	All MRI statistical features	Distribution analysis	StandardScaler normalization applied	Improve model convergence and stability

EDA Results

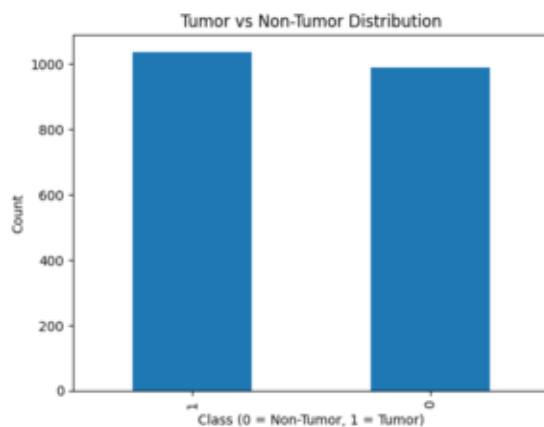
Exploratory Data Analysis (EDA) was performed to:

- understand MRI feature distributions,
- evaluate class balance,
- identify outliers,
- analyze feature correlations,
- and assess tumor-related MRI patterns.

Several important insights were identified during EDA.

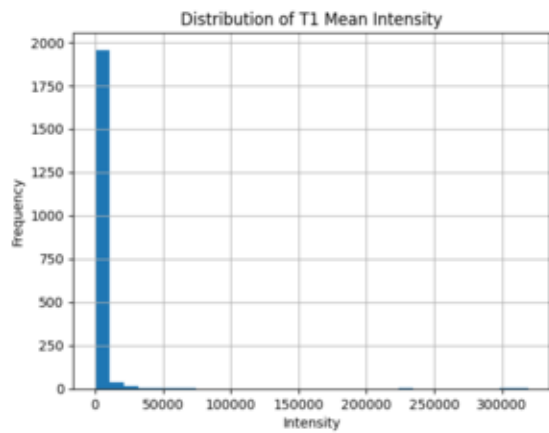
Tumor vs Non-Tumor distribution

Figure 1

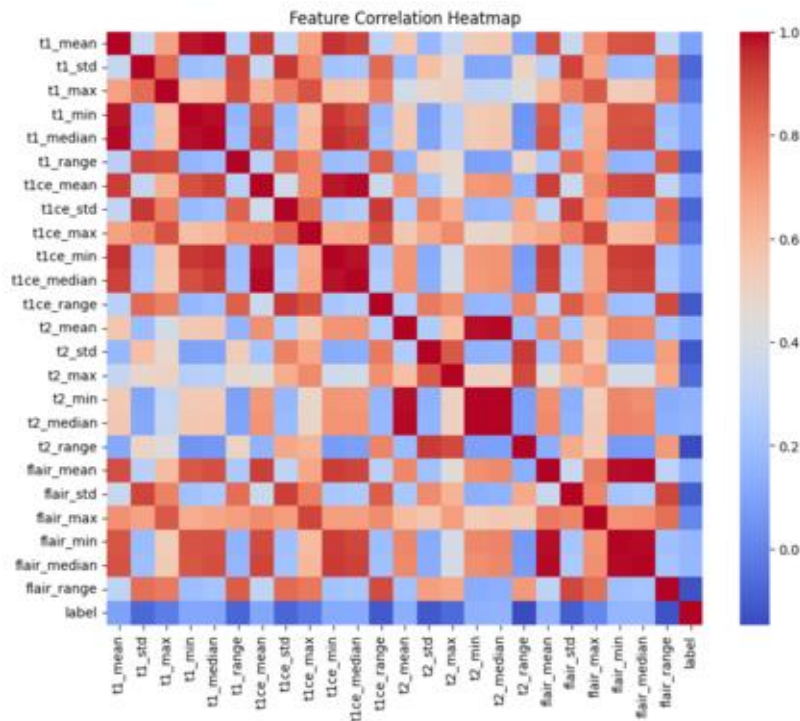


T1 mean intensity distribution

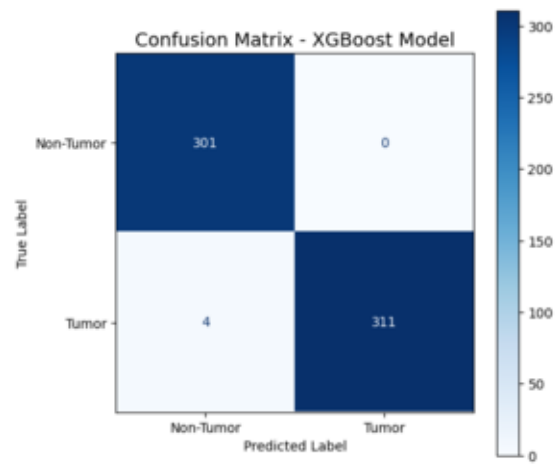
Figure 2



Correlation heatmap
Figure 3

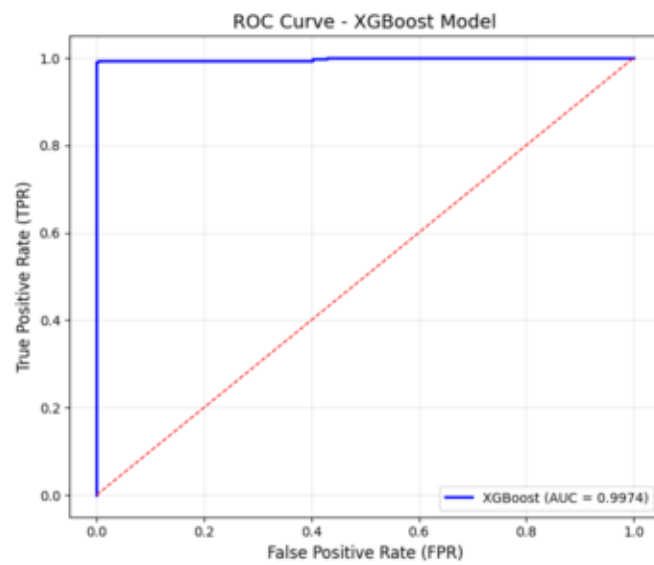


Confusion matrix
Figure 4



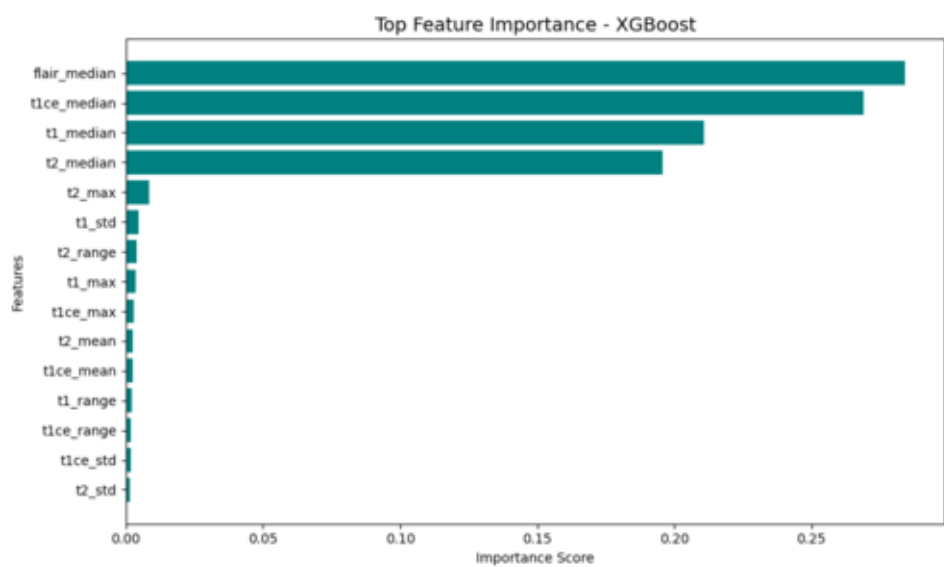
ROC curve

Figure 5



Feature importance plot

Figure 6



ML vs DL comparison
Figure 7

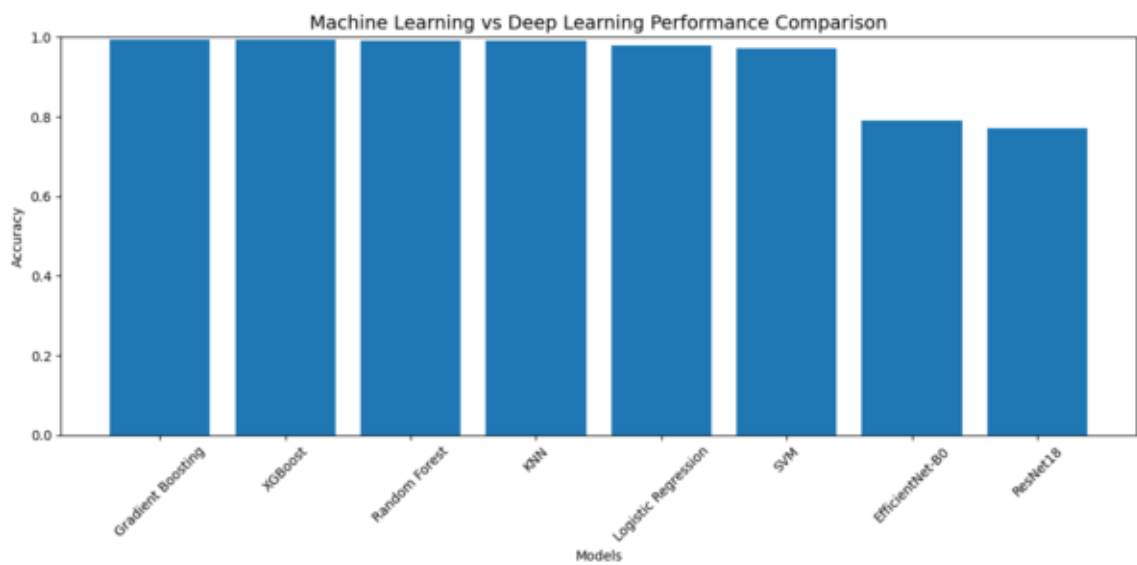


Table E

EDA Insight Summary

Figure/Table Reference	What it Shows	Key Insight	Why it Matters (Link to RQ/objective)	Decision/Next Step
Figure 1	Tumor vs Non-Tumor distribution	Dataset is relatively balanced	Reduces classification bias	Proceed with standard classification models
Figure 2	T1 mean intensity distribution	MRI intensity values contain extreme ranges	Indicates tumor heterogeneity	Apply normalization and outlier analysis
Figure 3	Correlation heatmap	Several MRI features are strongly correlated	Multi-modal MRI provides complementary information	Perform feature selection
Figure 4	Confusion matrix	Very low misclassification rate	Strong classification capability	Validate with GroupKFold
Figure 5	ROC curve	Excellent separability between classes	High predictive performance	Support hypothesis validation
Figure 6	Feature importance plot	FLAIR and T1CE features are highly informative	Certain MRI modalities contribute more strongly	Prioritize important modalities
Figure 7	ML vs DL comparison	Machine learning outperformed deep learning	Engineered features highly effective	Focus on ensemble ML models

Modelling

Choice of Models with Justification

Multiple machine learning and deep learning models were selected to evaluate glioma tumor classification performance.

Model 1: Logistic Regression

Used as a baseline linear classification model for comparison with advanced algorithms.

Model 2: Support Vector Machine (SVM)

Selected because SVM performs effectively on high-dimensional medical imaging datasets and can learn non-linear boundaries.

Model 3: K-Nearest Neighbors (KNN)

Used to evaluate similarity-based MRI feature classification.

Model 4: Random Forest

Chosen because ensemble tree methods reduce overfitting and capture non-linear MRI feature interactions.

Model 5: Gradient Boosting

Selected because boosting improves weak learner performance sequentially and achieves strong predictive accuracy.

Model 6: XGBoost

Selected due to:

- regularization capability,
- robustness,
- computational efficiency,
- and excellent structured data performance.

Model 7: ResNet18

Used as a transfer learning deep learning model capable of learning hierarchical MRI image features.

Model 8: EfficientNet-B0

Selected because it provides strong image classification capability with fewer trainable parameters and improved computational efficiency.

Features Included and Feature Engineering

The study used engineered MRI statistical intensity-based features extracted from:

- T1,
- T1CE,
- T2,
- and FLAIR MRI modalities.

A total of 24 statistical MRI features were extracted.

Feature engineering included:

- mean intensity,
- standard deviation,
- maximum intensity,
- minimum intensity,
- median intensity,
- and intensity range.

Table F**Feature Set and Usage**

Feature	Original/Engineered	Type	Reason for Inclusion	Used in Model(s)
t1_mean	Engineered	Numerical	Anatomical intensity representation	All ML models
t1ce_mean	Engineered	Numerical	Contrast-enhanced tumor detection	All ML models
t2_mean	Engineered	Numerical	Fluid-sensitive abnormalities	All ML models
flair_mean	Engineered	Numerical	Edema-sensitive feature	All ML models
t1ce_range	Engineered	Numerical	Tumor heterogeneity measurement	Ensemble models
flair_range	Engineered	Numerical	MRI variability representation	Ensemble models
MRI image slices	Original	Image data	Automatic visual feature learning	ResNet18, EfficientNet-B0

Table G**Evaluation metrics and formulae**

Metric	Formula	Interpretation	Used For
Accuracy	$(TP + TN) / (TP + TN + FP + FN)$	Measures the overall proportion of correctly classified tumor and non-tumor samples	Classification
Precision	$TP / (TP + FP)$	Measures how many predicted tumor cases were actually tumors	Classification
Recall (Sensitivity)	$TP / (TP + FN)$	Measures the ability of the model to correctly identify actual tumor cases	Classification
F1-score	$2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$	Harmonic mean of precision and recall; balances false positives and false	Classification

		negatives	
ROC-AUC	Area Under the ROC Curve	Measures the model's ability to distinguish between tumor and non-tumor classes	Classification

Where:

- **TP** = True Positives
- **TN** = True Negatives
- **FP** = False Positives
- **FN** = False Negatives

Research Questions and Hypotheses

Research Question 1 (RQ1)

How effectively can machine learning models classify glioma tumors using statistical intensity-based features extracted from multi-modal MRI scans?

Null Hypothesis (H_{01})

Machine learning models trained on engineered multi-modal MRI features will not achieve high glioma classification performance with accuracy and ROC-AUC scores greater than 95%.

Alternative Hypothesis (H1₁)

Machine learning models trained on engineered multi-modal MRI features will achieve high glioma classification performance with accuracy and ROC-AUC scores greater than 95%.

Research Question 2 (RQ2)

How accurately can deep learning transfer learning models classify glioma tumors directly from MRI image slices without manual feature extraction?

Null Hypothesis (H0₂)

Deep learning transfer learning models such as ResNet18 and EfficientNet-B0 will not achieve classification accuracy greater than 75% using MRI image slices.

Alternative Hypothesis (H1₂)

Deep learning transfer learning models such as ResNet18 and EfficientNet-B0 will achieve classification accuracy greater than 75% using MRI image slices.

Research Question 3 (RQ3)

Does integrating multiple MRI modalities improve glioma tumor classification performance compared to using a single MRI modality?

Null Hypothesis (H0₃)

Multi-modal MRI feature integration will not produce better classification performance than single-modality MRI analysis.

Alternative Hypothesis (H1₃)

Multi-modal MRI feature integration will produce better classification performance than single-modality MRI analysis.

Research Question 4 (RQ4)

Which approach provides better glioma tumor classification performance: traditional machine learning models using engineered MRI features or deep learning models trained directly on MRI images?

Null Hypothesis (H0₄)

Machine learning models using engineered MRI statistical features will not outperform deep learning transfer learning models for glioma tumor classification.

Alternative Hypothesis (H1₄)

Machine learning models using engineered MRI statistical features will outperform deep learning transfer learning models for glioma tumor classification.

Preliminary Results

Preliminary experiments were conducted to evaluate the effectiveness of machine learning and deep learning models for glioma tumor classification using multi-modal MRI data. The analysis included engineered MRI statistical features, transfer learning-based deep learning models, and comparative evaluation between single-modality and multi-modal MRI approaches. Performance was evaluated using accuracy, precision, recall, F1-score, ROC-AUC, confusion matrices, and validation analysis.

Preliminary Findings (by Research Question)**Research Question 1 (RQ1)****RQ1**

How effectively can machine learning models classify glioma tumors using statistical intensity-based features extracted from multi-modal MRI scans?

Null Hypothesis (H0₁)

Machine learning models trained on engineered multi-modal MRI statistical features will not achieve glioma classification performance with accuracy and ROC-AUC scores greater than 95%.

Alternative Hypothesis (H1₁)

Machine learning models trained on engineered multi-modal MRI statistical features will achieve glioma classification performance with accuracy and ROC-AUC scores greater than 95%.

Preliminary Results

The preliminary machine learning results demonstrated exceptionally high glioma tumor classification performance using engineered MRI statistical features extracted from T1, T1CE, T2, and FLAIR modalities. As shown in Figure 7 and Table F, ensemble-based machine learning models achieved the strongest predictive performance.

Gradient Boosting and XGBoost achieved the highest classification accuracy of **99.35%**, with ROC-AUC scores above **99.7%**. Random Forest and KNN also achieved strong performance with accuracies above **99%**.

The best-performing model, Gradient Boosting, achieved:

- Accuracy = 0.9935
- ROC-AUC = 0.9974

- Precision = 1.0000
- Recall = 0.9873
- F1-score = 0.9936

These findings indicate that engineered multi-modal MRI statistical features are highly informative for glioma tumor classification.

The results strongly support the effectiveness of ensemble learning methods for structured MRI feature datasets and confirm that machine learning models can accurately distinguish tumor and non-tumor MRI slices.

Hypothesis Evaluation

Since the best-performing machine learning models achieved both:

- Accuracy > 95%
- ROC-AUC > 95%

the null hypothesis (H_{01}) is rejected and the alternative hypothesis (H_{11}) is supported.

Research Question 2 (RQ2)

RQ2

How accurately can deep learning transfer learning models classify glioma tumors directly from MRI image slices without manual feature extraction?

Null Hypothesis (H_{02})

Deep learning transfer learning models such as ResNet18 and EfficientNet-B0 will not achieve classification accuracy greater than 75% using MRI image slices.

Alternative Hypothesis (H_{12})

Deep learning transfer learning models such as ResNet18 and EfficientNet-B0 will achieve classification accuracy greater than 75% using MRI image slices.

Preliminary Results

Deep learning transfer learning models demonstrated moderate but effective classification performance on MRI image slices. EfficientNet-B0 achieved the highest deep learning validation accuracy of **79.14%**, while ResNet18 achieved **77.05%** validation accuracy.

EfficientNet-B0 also achieved:

- Precision = 0.7299
- Recall = 0.8850
- F1-score = 0.8000
- ROC-AUC = 0.8365

The higher recall values indicate that the transfer learning models successfully identified a large proportion of tumor cases.

However, deep learning performance remained lower than machine learning performance, possibly due to:

- limited dataset size,
- MRI image complexity,
- and computational constraints.

The findings confirm that transfer learning approaches can classify glioma MRI images effectively without manual feature extraction, although additional data and training optimization may further improve performance.

Hypothesis Evaluation

Since EfficientNet-B0 achieved:

- Accuracy = 79.14%

which is greater than the proposed threshold of 75%, the null hypothesis (H_{02}) is rejected and the alternative hypothesis (H_{12}) is supported.

Research Question 3 (RQ3)

RQ3

Does integrating multiple MRI modalities improve glioma tumor classification performance compared to using a single MRI modality?

Null Hypothesis (H_{03})

Multi-modal MRI feature integration will not produce better classification performance than single-modality MRI analysis.

Alternative Hypothesis (H_{13})

Multi-modal MRI feature integration will produce better classification performance than single-modality MRI analysis.

Preliminary Results

The comparative experiments showed that multi-modal MRI integration improved classification performance compared to single-modality MRI analysis.

As shown in Table X:

- T1-only classification achieved 99.18% accuracy
- Multi-modal MRI classification achieved 99.51% accuracy

The multi-modal model also achieved:

- Precision = 0.9968
- Recall = 0.9936
- F1-score = 0.9952
- ROC-AUC = 0.9992

These findings demonstrate that combining multiple MRI modalities provides complementary tumor information that improves glioma classification capability.

The results strongly support the use of multi-modal MRI integration for accurate tumor detection and classification.

Hypothesis Evaluation

Since the multi-modal MRI model achieved higher performance than the T1-only model, the null hypothesis (H_{03}) is rejected and the alternative hypothesis (H_{13}) is supported.

Research Question 4 (RQ4)

RQ4

Which approach provides better glioma tumor classification performance: traditional machine learning models using engineered MRI features or deep learning models trained directly on MRI images?

Null Hypothesis (H_{04})

Machine learning models using engineered MRI statistical features will not outperform deep learning transfer learning models for glioma tumor classification.

Alternative Hypothesis (H_{14})

Machine learning models using engineered MRI statistical features will outperform deep learning transfer learning models for glioma tumor classification.

Preliminary Results

The comparative evaluation demonstrated that machine learning models significantly outperformed deep learning transfer learning models on the current MRI dataset.

The best machine learning models:

- Gradient Boosting and XGBoost,
- achieved classification accuracies above 99%.

In contrast:

- EfficientNet-B0 achieved 79.14%
- ResNet18 achieved 77.05%

The superior performance of machine learning models suggests that carefully engineered statistical MRI intensity features contain highly discriminative tumor information for glioma classification.

The results also indicate that:

- ensemble machine learning approaches remain highly effective for structured MRI feature datasets,
- while deep learning approaches may require larger datasets and longer training for improved generalization.

Hypothesis Evaluation

Since machine learning models achieved substantially higher classification accuracy than deep learning models, the null hypothesis (H0₄) is rejected and the alternative hypothesis (H1₄) is supported.

Table F**Preliminary Model Performance Comparison**

Model	Feaure Set	Validation Approach	Accuracy	ROC-AUC / F1	Key Takeaway
Gradient Boosting	Multi-modal MRI statistical features	Train-test split + GroupKFold	0.9935	ROC-AUC = 0.9974	Best overall classification performance
XGBoost	Multi-modal MRI statistical features	Train-test split	0.9935	ROC-AUC = 0.9973	Strong ensemble learning performance
Random Forest	Multi-modal MRI statistical features	Train-test split	0.9919	ROC-AUC = 0.9990	Excellent non-linear feature learning
KNN	Multi-modal	Train-test	0.9919	F1 = 0.9920	Strong

	MRI statistical features	split			similarity- based classification
Logistic Regression	Multi-modal MRI statistical features	Train-test split	0.9789	F1 = 0.9789	Effective baseline linear classifier
SVM	Multi-modal MRI statistical features	Train-test split	0.9708	F1 = 0.9706	Strong high- dimensional classification
EfficientNet- B0	MRI image slices	Validation split	0.7914	F1 = 0.8000	Best deep learning model
ResNet18	MRI image slices	Validation split	0.7705	F1 = 0.7730	Moderate deep learning performance

Preliminary Model Performance Comparison

Model	Accuracy	ROC-AUC	Precision	Recall	F1-Score
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Gradient Boosting	0.9935	0.9974	1.0000	0.9873	0.9936
XGBoost	0.9935	0.9973	1.0000	0.9873	0.9936
Random Forest	0.9919	0.9990	1.0000	0.9841	0.9920
KNN	0.9919	0.9936	1.0000	0.9841	0.9920
Logistic Regression	0.9789	0.9977	1.0000	0.9587	0.9789
SVM	0.9708	0.9879	1.0000	0.9429	0.9706
EfficientNet-B0	0.7914	0.8365	0.7299	0.8850	0.8000
ResNet18	0.7705	0.8137	0.7242	0.8289	0.7730

Performance Analysis

As shown in Figure X, machine learning models consistently achieved higher classification accuracy than deep learning transfer learning models. Ensemble learning approaches such as Gradient Boosting, XGBoost, and Random Forest demonstrated superior classification capability for engineered MRI statistical features.

Gradient Boosting achieved the highest overall classification performance with:

- Accuracy = 99.35%
- ROC-AUC = 99.74%
- Precision = 100%
- Recall = 98.73%
- F1-score = 99.36%

The confusion matrix and ROC curve analysis further confirmed the strong discriminative capability of the Gradient Boosting and XGBoost models.

Deep learning transfer learning models achieved comparatively lower performance, likely due to:

- limited MRI image dataset size,
- complexity of MRI tumor patterns,
- and reduced training epochs.

The comparative analysis also demonstrated that:

- multi-modal MRI integration improved classification capability,
- feature engineering remained highly effective,
- and ensemble machine learning methods were highly suitable for glioma tumor classification

using structured MRI intensity-based features.

Interim Limitations and Risks

Several limitations and risks were identified during the interim phase of the project. These limitations may affect model generalization, computational efficiency, and final deployment performance.

1. Limited Deep Learning Dataset Size

The deep learning models were trained using a relatively limited number of MRI image slices. Transfer learning models such as ResNet18 and EfficientNet-B0 generally perform better with larger image datasets.

Mitigation Planned:

Additional MRI image augmentation and extended training strategies will be explored to improve deep learning generalization performance.

2. MRI Intensity Variability

MRI scans obtained from different patients may contain variability in:

- intensity distribution,
- imaging quality,
- and scanner characteristics.

This variability can affect feature consistency and model robustness.

Mitigation Planned:

Additional normalization and preprocessing techniques will be evaluated to improve feature standardization across MRI samples.

3. Potential Overfitting in Ensemble Models

Machine learning ensemble methods such as Gradient Boosting and XGBoost achieved extremely high classification accuracy, which may indicate a potential risk of overfitting.

Mitigation Planned:

Additional validation techniques such as:

- cross-validation,
- patient-wise GroupKFold validation,
- and hyperparameter optimization

will be further performed to ensure model generalizability.

4. Limited Hyperparameter Optimization

The current experiments used preliminary hyperparameter configurations due to time and computational constraints.

Mitigation Planned:

Grid Search and Randomized Search optimization techniques will be applied during the final phase to improve model tuning and performance stability.

5. Computational Resource Constraints

Deep learning model training required significant GPU computational resources and training time.

Mitigation Planned:

Model optimization and efficient training configurations will be implemented to reduce computational overhead.

6. Lack of External Validation Dataset

The current study used a single publicly available glioma MRI dataset. External clinical validation has not yet been performed.

Mitigation Planned:

Additional validation using independent MRI datasets will be considered if accessible during the final project phase.

Next Steps (for Final Report)

- Perform advanced hyperparameter tuning for machine learning and deep learning models
- Conduct additional cross-validation experiments for robustness evaluation
- Improve deep learning performance using data augmentation and training optimization

- Perform detailed feature importance and model interpretability analysis
- Finalize statistical hypothesis testing and validation
- Expand comparative analysis between single-modality and multi-modal MRI classification
- Conduct additional error analysis using confusion matrices and ROC analysis
- Organize and finalize GitHub repository structure and documentation
- Prepare final conclusions, recommendations, and future research directions

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